

Nutrition Throughout The Life Cycle

CHAPTER 8

ADOLESCENT NUTRITION

OBJECTIVES

- **At the end of this chapter you will be able to:**
 - Identify the physical growth and development that occurs during adolescence.
 - Discuss the common eating behaviours and food intake habits of adolescents.
 - Identify the factors that influence eating behavior and food choice of adolescents.
 - Discuss the energy and nutrient requirements of adolescents.

Introduction

- **Adolescence** is defined as:
 - the period between the onset of puberty and the cessation of physical growth; *i.e.* the period of life between 10 to 19 years (WHO).
- It is a time of profound physical, emotional, social, and cognitive changes during which a child develops into an adult.
- Biologically, adolescence is the time of life when growth is completed and individuals become sexually mature.

Physical Growth and Development

- The biological changes that occur during puberty include **sexual maturation, increases in height and weight, accumulation of skeletal mass**, and changes in body composition.
- The sequence of these events is consistent; the age of onset, duration, and tempo of these events vary between individuals.
 - For example a 14-year-old male who has already experienced rapid linear growth and muscular development will have different energy and nutrient needs than a 14-year-old male peer who has not yet entered puberty.
 - These variations directly affect the nutrition requirements of adolescents.

Changes in Weight, Body Composition, and Skeletal Mass

- As much as 50% of ideal adult body weight is gained during adolescence.
- Body composition changes dramatically among females during puberty.
- Adolescent **body fat levels peak** among females between the ages of 15 and 16 years.
- Almost **half of adult peak bone mass** is accrued during adolescence.

Table 14.2 Psychosocial processes and the substages of adolescent development

Substage	Emotional and Social Development	Cognitive Development
Early adolescence	• Alterations in body image secondary to dramatic changes in body shape and size • Increased awareness of sexuality • Strong need for social acceptance by peers • Strong sense of impulsivity	• Concrete thinking processes are dominant, often with limited abstract thought capacity
Middle adolescence	• Development of greater autonomy from parents and family • Continued need for peer acceptance, which may lead to risk-taking behaviors • Increased opportunities for employment outside of home, resulting in more decision making and the beginning of economic independence • Increased awareness of moral and social issues	• Development of abstract reasoning continues • May revert to concrete thinking when under stress
Late adolescence	• Further development of personal set of morals and values • Increased impulse control • Greater social, emotional, and economic independence from family • Reduced need for peer acceptance • Development of personal and vocational goals	• Abstract thought capacity fully develops

SOURCE: From G. M. Ingersoll, "Psychological and Social Development," in Textbook of Adolescent Medicine. Copyright © 1992 Saunders. Reprinted with permission.

Psychosocial Development

- Adolescents become under pressure:
 - ✓ To drink alcohol, to smoke and to experiment with drugs.
 - ✓ Show more independence from parents *i.e.* spend less time with parents and more time with friends.
 - ✓ Risk taking; trying out their new decision-making skills, and
 - ✓ Develop realistic assessments of themselves, other people, and the world.
- They are not usually interested in food and nutrition except as part of fashion.

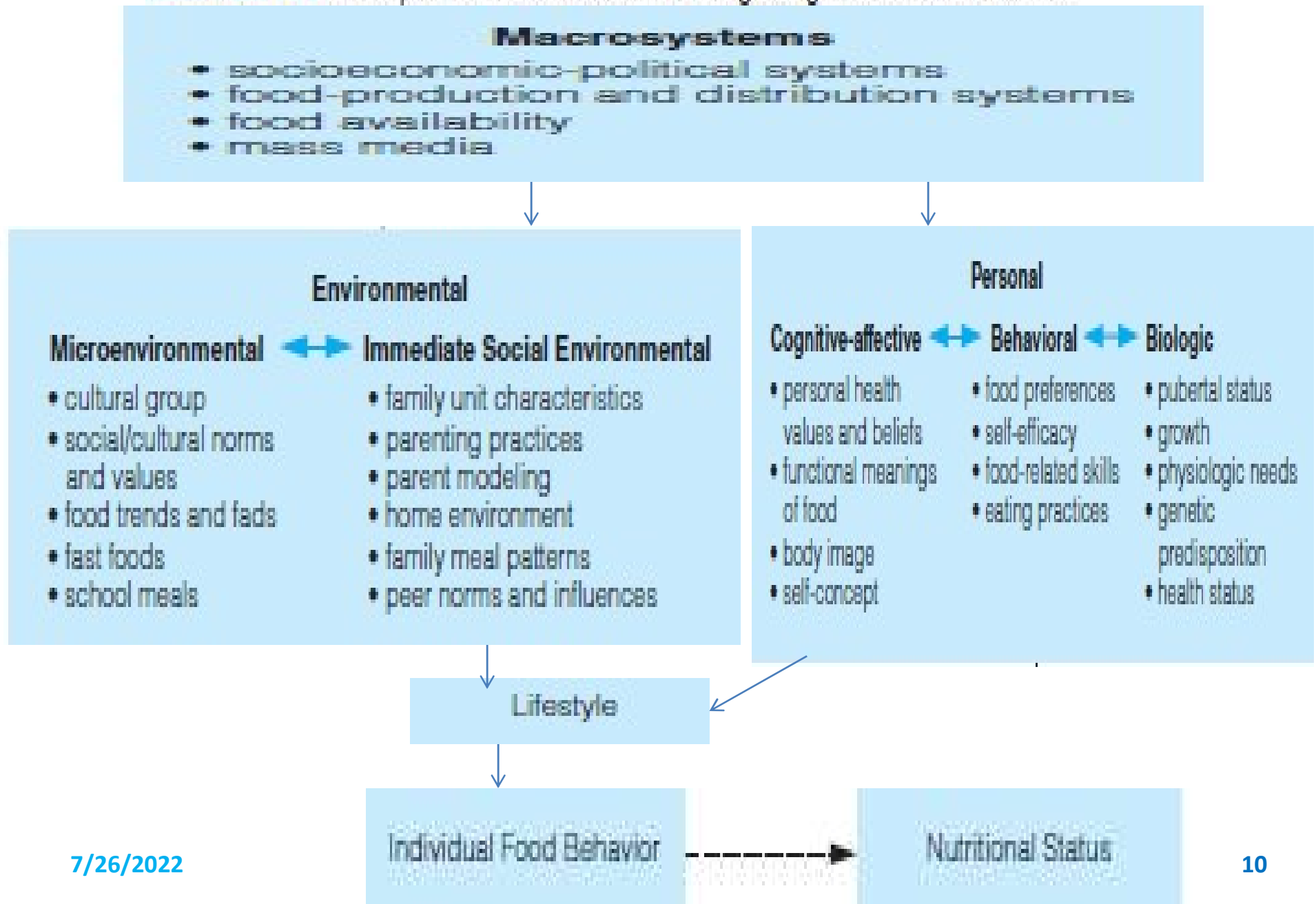
Health and Eating Related Behaviors During Adolescence

- Eating patterns and behaviors of adolescents are influenced by many factors, like; peer influences, parental modeling, food availability, food preferences, cost, convenience, cultural beliefs, socioeconomic status, racial/ethnic background, mass media, and body image.
- There are three interacting factors that influence adolescent's eating behaviors: personal or individual, environmental, and macrosystem.

Factors that influence eating behavior of Adolescents

- ❖ **Personal factors:** include attitudes, beliefs, food preferences, self-efficacy and biological changes.
- ❖ **Environmental factors:** include the immediate social environment such as family, friends/peers, and other factors such as school, fast-food and socio-cultural norms.
- ❖ **Macrosystem factors:** include food availability, food production and distribution systems, socio-economic & political systems, mass media and advertising.

Illustration 14.4 Conceptual model for factors influencing eating behavior of adolescents.



Common Eating Behaviours of Adolescents

1. Snacking

- Snacking is common among adolescents.
- Snacks account for **25-33% of** daily energy intakes among adolescents.
- Food choices made by adolescents while snacking tend to be high in **sugar, sodium, and fat**, while relatively low in **vitamins and minerals**.
- **Soft drinks** are the most commonly chosen snacks for adolescent females; high consumption of soft drinks increases the risk for bone fractures over lifetime.
 - ✓ Over time, these changes in nutrient intake could lead to an increased risk for chronic health problems such as osteoporosis and obesity.

Common Eating Behaviours...

2. Inactivity and Slimming

- Many adolescents have low energy intakes.
- There are three important consequences of low energy intakes in adolescent girls:
 1. They may not achieve their requirement for **iron**.
 2. They may not achieve their requirement for **calcium**, (*i.e.* low peak bone mass which is a risk factor for osteoporosis in later life).
 3. They may not achieve their **full genetic potential in height**, especially if they use cigarette smoking as a way of controlling food intake.

Common Eating Behaviours...

3. Skipping Meals

- Meal skipping is common among adolescents, especially during middle and late adolescence.
- Breakfast is the most commonly skipped meal and is attributed to **lack of time, desire to sleep longer in the morning, lack of appetite and dieting to lose weight.**
- Skipping breakfast may affect **concentration, learning, school performance and leads to having lower total daily energy, protein, vitamin and mineral intakes.**

Common Eating Behaviours...

4. Eating Out

- During adolescence, teens spend less time with family and more time with their **peer group** and eat more foods at fast food restaurants.
- As teens become more independent, eating away from home increases. The reasons for this may be:
 - ✓ Fast foods are relatively **inexpensive** and offer socially acceptable choices,
 - ✓ Fast foods can be eaten outside the restaurant,
 - ✓ The service is fast, and
 - ✓ To increase the social value/ to get friends.

Vegetarian Diets

Table 14.3 Types of vegetarian diets and foods excluded

Type of Vegetarian Diet	Foods Excluded
Semi- or partial-vegetarian	Red meat
Lacto-ovo-vegetarian	Meat, poultry, fish, seafood
Lacto-vegetarian	Meat, poultry, fish, seafood, eggs
Vegan (total vegetarian)	Meat, poultry, fish, seafood, eggs, dairy products (may exclude honey)
Macrobiotic	Meat, poultry, eggs, dairy, seafood, fish (fish may be included in the diets of some macrobiotic vegetarians)

source: Reprinted with permission. Haddad E, Johnston P. Vegetarian Diets and Pregnant Teens. In: Story M, Stang J, eds. Nutrition and the Pregnant Adolescent: A Practical Reference Guide. Minneapolis, MN: Center for Leadership, Education, and Training in Maternal and Child Nutrition, University of Minnesota; 2000.

Energy and Nutrient Requirements of Adolescents

- Increases in lean body mass, skeletal mass, and body fat that occur during puberty result in energy and nutrient needs that exceed those at any other point in life.
- Energy and nutrient requirements of adolescence correspond with the degree of physical and maturation that has taken place.
- Adolescents consume inadequate vitamins (folate; vitamins A, B6, C & E) and minerals (iron, zinc, magnesium, phosphorus & calcium).
 - ✓ Dietary fiber intake among adolescents is also low.
 - ✓ Total & saturated fats, cholesterol, sodium, and added sugar consumed by many teens *exceed* recommendations.

Energy

- Energy needs of adolescents are influenced by activity level, basal metabolic rate, and there is increased requirements of energy to support pubertal growth and development.
- Adolescent males have significantly higher caloric requirements than females, since they experience greater increases in height, weight, and lean body mass.
- Caloric needs of older adolescents who have completed puberty and are less active may be significantly lower than those of younger, active, still-growing adolescents.
- Physical growth and development during puberty is sensitive to energy and nutrient intakes.
 - When energy intakes fail to meet requirements, linear growth may be retarded and sexual maturation may be delayed.

Carbohydrates

- Carbohydrates provide the body's primary source of dietary energy.
- The recommended intake of carbohydrate among teens is **130 g/day which covers 45–65%** of daily energy needs.
- Dietary recommendations suggest that 50% or more of total daily calories should come from carbohydrate, with no more than **10-25% of calories derived from sweeteners**, such as sucrose and high fructose corn syrup.
 - Fruits, vegetables, whole grains, and legumes are the main source of both carbohydrates and dietary fiber.

Dietary Fiber

- Dietary fiber is important for normal bowel function.
 - It reduces serum cholesterol levels (prevents coronary artery disease) and also reduce the risk of obesity and certain cancers.
 - It moderates blood sugar levels and prevents type 2 diabetes mellitus).
- recommendation
 - females 26 g/day,
 - for males <14 years of age 31 g/day, and for older adolescent males 38 g/day.

Protein

- Protein needs of adolescents are influenced by the amount of protein required for
 - maintenance of existing lean body mass,
 - allowances for the amount required to accrue additional lean body mass during the adolescent growth spurt.
- Recommended---0.85 g/kg body weight/day
- Protein requirements are highest for
 - females at 11 to 14 years and for
 - males at 15 to 18 years, when growth is at its peak.

Protein...

- As with energy, growth is affected by protein intakes.
 - reductions in linear growth,
 - delays in sexual maturation, and
 - reduced accumulation of lean body mass

Fat

- The human body requires dietary fat and essential fatty acids for normal growth and development.
- The DRIs recommend specific intake of Linoleic and alpha-linolenic acid to support optimal growth and development.
- 30% of daily calories should be obtained from total fat and not more than 10% from saturated fat.

Minerals

Calcium

- ◆ Calcium is the main constituent of bone mass.
- ◆ Calcium needs during adolescence are greater than they are in either childhood or adulthood because of the dramatic increase in skeletal growth (about 45% of peak bone mass is attained during adolescence).
- ◆ Adequate calcium intake is important for the development of dense bone mass and the reduction of the lifetime risk of fractures and osteoporosis.

Calcium...

- By age 17, adolescents have attained approximately 90% of their adult bone mass.
- Calcium needs and absorption rates are higher during **adolescence** than any other time except infancy.
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- ▶ The DRI of calcium for 9 to 18 year olds is 1300 mg/day.

Calcium...

- The consumption of soft drinks by adolescents may displace the consumption of more nutrient-dense beverages, including milk and fortified juices.
- Females who restrict calories to control their weight are particularly at high risk for inadequate calcium intakes.

Iron

- The rapid rate of linear growth, the increase in blood volume, and the onset of menarche during adolescence increase a teen's need for iron.
- Iron needs of an adolescent will be highest during the adolescent growth spurt in males, and after menarche in females.
 - The onset of menstruation imposes additional iron needs for girls.

Iron...

- Iron deficiency occurs more frequently in all adolescents and iron-deficiency anemia occurs almost exclusively in females.
- The RDA for iron is 8 mg/day for 9-13 year olds, 11 mg/day for males ages 14-18 and 15 mg/day for females ages 14-18.

Zinc

- ▶ Zinc is important during adolescence because of its role in **growth and sexual maturation**.
- ▶ Males who are zinc deficient experience growth failure and delayed sexual development.
- ▶ Serum zinc levels decline in response to the rapid growth and hormonal changes that occur during adolescence.
- ▶ The RDA for zinc for males and females ages 9-13 is 8 mg/day.
- ▶ For males and females ages 14-18, the RDA is 11 mg/day and 9 mg/day, respectively.

Zinc...

- ▶ Zinc is naturally abundant in red meats, shellfish, whole grains and fortified breakfast cereals.
- ▶ Absorption of zinc can be inhibited by **indigestible fibers found in plant-based foods**.
- ▶ Zinc and iron compete for absorption, so elevated intakes of one can reduce the absorption of the other.
 - ▶ **Adolescents who take iron supplements may be at increased risk of developing mild zinc deficiency if iron intake is over twice as high as that of zinc.**
- ▶ **Vegans**, and teens who don't consume animal-derived products are at highest risk for low intakes of zinc.

Vitamins

Vitamin A

- Vitamin A is important for normal vision, reproduction, growth and immune function.
- To ensure adequate body stores of vitamin A, boys and girls aged 9-13 should consume 600 µg/day, females & males aged 14-18 years need 700 & 900 µg/day.
- Dietary sources of vitamin A are meat, milk, cheese, ready-to-eat cereal, carrots, margarine, tomatoes, spinach, sweet potato, yellow fruits and green leafy vegetables.

Vitamin D

- Vitamin D can be synthesized by the body.
- It facilitates intestinal absorption of calcium and phosphorus.
- The AI for vitamin D among adolescents is 200 IU (5 µg) per day; the upper limit is 2000 IU (50 µg) per day.
- The American Academy of Pediatrics (AAP) recommended that all adolescents to consume at least 400 IU of vitamin D per day through dietary sources.
 - ▶ Vitamin D is naturally found in very few foods like, fatty fish, fish oils & egg yolks.

Vitamin E

- Vitamin E is a known antioxidant, which become increasingly important as body mass expands during adolescence.
- The RDA for vitamin E for 9-13 year olds is 11 mg/day and 15mg/day for 14-18 year olds.
 - ▶ Fortified breakfast cereals, margarine, mayonnaise and nuts/seeds are good sources of vitamin E.

Folate

- ◆ Adolescents have increased requirements for folate during puberty.
- ◆ Poor folate status among adolescent females have an issue related to reproduction.
- ◆ Teens who skip breakfast or do not commonly consume orange juice and ready-to-eat cereals are at an increased risk for having a low consumption of folate.
- ◆ The RDA for folate is 300µg/day for 9-13 year olds, and 400µg/day for 14-18 year olds.
 - ◆ Folic acid is added to fortified cereals, breads, and other refined grain products in the form of folate.

Vitamin C

- Vitamin C plays an important role during adolescent growth and development.
- Smoking increases the need for antioxidant within the body because it consumes vitamin C in oxidative reactions.
- The RDA for vitamin C is 45 mg/day for 9-13 year olds, 75 mg/day for males aged 14-18 and 65 mg/day for females aged 14-18; **smokers +35 mg/day**.
- Recommended level of vitamin C intake is higher among smokers; since smoking increases oxidative stress and metabolic turnover of vitamin C.
 - ▶ Food sources: orange and grapefruit juice, vegetables, ready-to-eat cereals, tomatoes, and white potatoes.

Factors that influence Adolescents' food choice

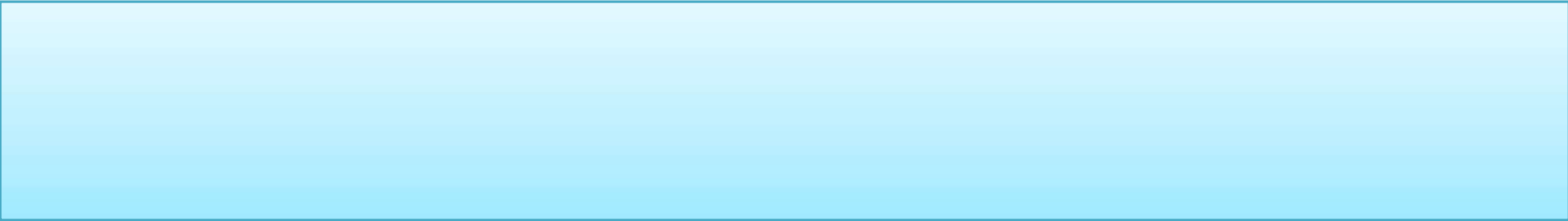
- Adolescents have many factors influencing their eating choices.
 - ✓ Eating out, busy lifestyle and limited incomes or allowances, fast food restaurants are a popular choice.
 - ✓ Media (advertisements) and family can also influence what teens eat.
- **Income levels**: household income and food sufficiency status.
- **Racial/Ethnic Groups**: higher intake of whole grains among white youth; lower intake of dark green leafy vegetables by Hispanics; higher whole milk consumption among Hispanics and blacks; lower total milk consumption among blacks; and higher intake of soda by whites.

Nutrition condition during Adolescence

- Chronic problem of low birth weight is more common in the offspring of adolescent mothers.
- Adolescence is a key time from a nutritional standpoint.
- Lack of adequate quality and quantity of food is a prime cause of nutrition problems in adolescents.
- Chronic undernutrition (stunting) among young people delays growth and physical maturation, increases risk to pregnant mothers and their newborns, and decreases the capacity to work.
- Folate and iron deficiency are of particular concern in adolescence, along with other micronutrient deficiencies.

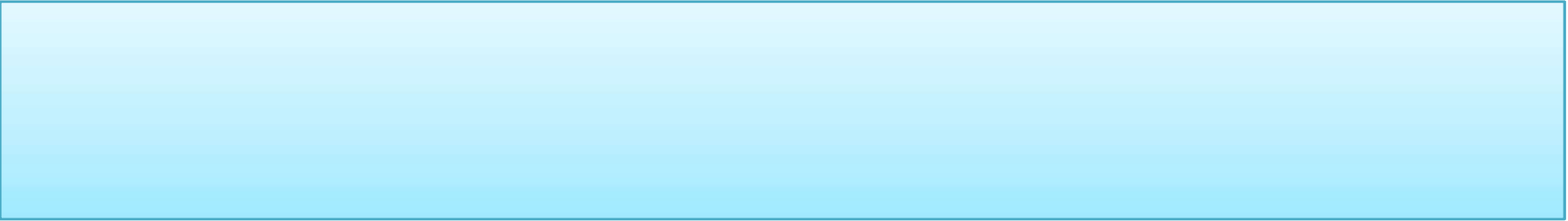
Nutrition condition during Adolescence...

- There is also a connection between poor maternal nutrition and greater risk of transmission of HIV from mothers to their infants.
- Many harmful eating habits begin in adolescence and can lead to diet-related chronic illness later in life.
- Adolescence offers an early chance to acquire knowledge about optimal nutrition and to contribute to the prevention or reduction of:
 - problems during pregnancy and;
 - diet-related illnesses.



Eating Disorders

- The common feature of the different eating disorders is the **abnormal eating behaviors**.
- Adolescents tend to be thin. Fear of gaining weight may lead to overly restrictive eating habits.
- Anorexia nervosa and bulimia nervosa are common eating disorders of adolescents.
- People with anorexia or bulimia nervosa view themselves as overweight while being abnormally underweight.

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- **Anorexia nervosa** is a form of self-starvation. restricting caloric intake due to fear of body weight
 - Weight loss
 - **Bulimia nervosa** is uncontrolled episodes of overeating and usually followed by self-induced vomiting, or any other action to control weight.
 - May have A normal or low body weight

Obesity

- **risk of adolescent obesity?**

- ✓ is the kind of food that adolescents eat, as they like junk food, which are very rich in fats, oil and cholesterol.
- ✓ sedentary lifestyle of adolescents.

Iron deficiency Anemia

- ✓ is one of the most common deficiency diseases among adolescents.
- ✓ This is due to low intake and absorption of iron,

THANK YOU!